

Deployment Specification

Student OS Documentation — File 15

Environments

Local development, staging/test and production should be logically separated.

Frontend

Production build, environment configuration, HTTPS domain, caching strategy and error monitoring.

Backend

Production server/runtime, environment variables, database connection, logging, health endpoint and graceful error handling.

Database

Managed/secured database where possible, migrations, backups, indexes and least-privilege credentials.

Environment Variables

Database URL, auth secrets, AI provider keys, storage credentials and other secrets must be configured outside source control.

CI/CD

Recommended pipeline: install → lint → test → build → migration/deploy steps → smoke checks.

Monitoring

Track application errors, API failures, latency, background jobs and notification failures without exposing sensitive data.

Rollback

Maintain deployable previous versions and reversible migration strategy for production incidents.

Production Checklist

HTTPS; secure cookies; CORS; secrets; database backup; migrations; monitoring; error handling; rate limiting; domain; final smoke tests.